

Note 5.3 Least Squares Problems

Least Squares Solutions of Overdetermined Systems

The least squares problem is to find a vector $\mathbf{x}$ such that $A\mathbf{x}$ is "closest" to $\mathbf{b}$ where A is a $m \times n$ matrix with $m > n$, that is, a **overdetermined system**.

The least square solution is to minimize the **residual**, that is,

$$|r(\mathbf{x})| = |\mathbf{b} - A\mathbf{x}|$$

is minimum.

Theorem 5.3.1

There **exists** a **unique** element $\mathbf{p} \in S$ where S is a subspace of $\mathbb{R}^m$ such that

$$\|\mathbf{b} - \mathbf{y}\| > \|\mathbf{b} - \mathbf{p}\|$$

where $\mathbf{b} \in \mathbb{R}^m$. That is, there exists a unique element $\mathbf{p}$ that is closest to $\mathbf{b}$.

And $\mathbf{p}$ is closest to $\mathbf{b}$ if and only if there **difference is the orthogonal complement** of S , that is, $\mathbf{b} - \mathbf{p} \in S^\perp$.

Here, in least square problems, $\mathbf{p} \in R(A)$, and $\mathbf{p} = A\hat{\mathbf{x}}$, the **projection** of $\mathbf{b}$ onto $R(A)$. And $r(\hat{\mathbf{x}})$ is perpendicular to $R(A)$.

Theorem 5.3.2

For $m \times n$ matrix A **with rank n** , the **normal equations**

$$A^T A \mathbf{x} = A^T \mathbf{b}$$

have a unique solution

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

And the projection vector $\mathbf{p} = A\hat{\mathbf{x}} = A(A^T A)^{-1} A^T \mathbf{b}$.